

MRI Brain Tissue Analysis Report

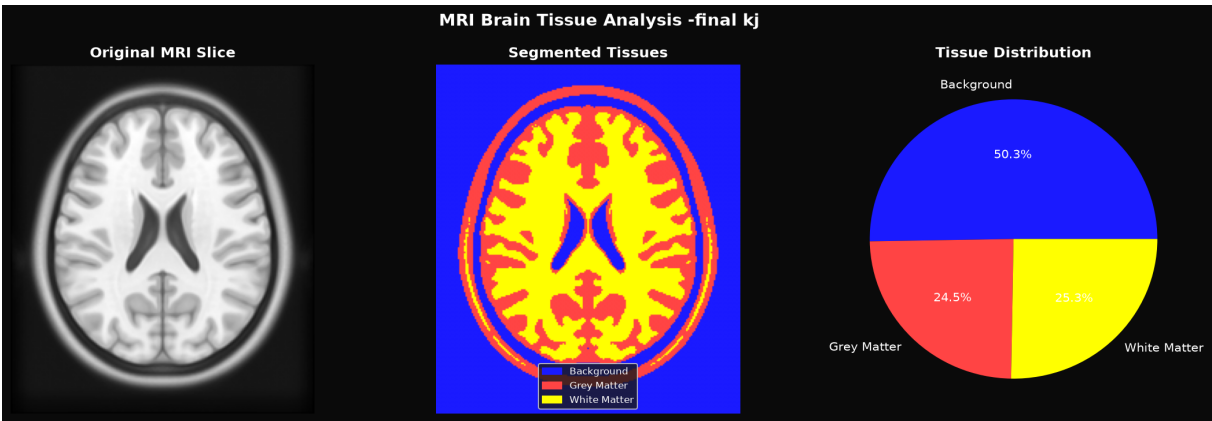
Generated: 19 June 2026, 12:16

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1. Introduction

This report presents an automated analysis of MRI brain tissue segmentation using the ICBM152 standard brain template. K-means clustering was applied to classify brain tissue into three categories: background, grey matter, and white matter.

2. Visualisation



3. Tissue Measurements

Tissue Type	Voxel Count	Percentage (%)	Avg Brightness
Background	23,075	50.27%	10.30
Grey Matter	11,226	24.46%	56.09
White Matter	11,600	25.27%	79.53

4. Key Findings

The analysis identified three distinct tissue regions. The dominant region is Background at 50.3% of the total slice area. Grey matter and white matter together represent the active brain tissue regions, with their ratio providing insight into brain structure and composition.

5. Methodology

Segmentation was performed using K-means clustering (k=3) on pixel intensity values of the axial mid-brain slice. Clusters were sorted by mean brightness to ensure consistent tissue labelling across different scans. All measurements are based on 2D slice analysis of the ICBM152 T1-weighted MRI template.